The Journal of Open Source Software

LAMMPS-GUI: A Cross-Platform Graphical Tool to Learn and Explore Molecular Dynamics with LAMMPS

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DOI: 10.xxxxxx/draft

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Submitted: 29 June 2026
Published: unpublished

Summary

LAMMPS is a widely used, massively parallel classical molecular dynamics (MD) engine for particle-based simulations at the atomic, mesoscopic, and continuum scales (Thompson et al., 2022). Implemented in C++ as a no-frills, console-mode application, it is highly portable and efficient, but learning to use it requires mastering separate – and often platform-specific – tools for editing inputs, plotting thermodynamic data, and visualizing atomic configurations *before* one can begin to learn MD itself.

LAMMPS-GUI is a cross-platform graphical application, written in C++17 using version 6 of the Qt framework (The Qt Company, 2026), that combines these tasks in one program: a syntax-highlighting input editor with auto-completion and documentation lookup, live execution with real-time output monitoring, interactive thermodynamic charts, a snapshot image creator, and a slide-show viewer. Rather than launching LAMMPS as an external process, it embeds the engine through the C-language library interface (FrantzDale et al., 2010), running the simulation in a concurrent worker thread so the interface stays responsive and the run can be monitored, plotted, visualized, and cleanly stopped. By mirroring the traditional "edit, run, observe, analyze" workflow, it lets users move freely to the command-line executable – essential once a workflow outgrows a personal computer. Pre-compiled packages for Windows, macOS, and Linux let most users start instantly, without compiling anything.

Statement of need

LAMMPS-GUI grew out of teaching LAMMPS at tutorials and workshops, where much of the available time went not to molecular dynamics but to installing and operating the surrounding tool chain – a text editor, a plotting program, and a molecular visualization package – most of which differ between Windows, macOS, and Linux. Earlier workarounds were either too complex to sustain or broke when Apple moved to the ARM architecture. As a single, self-contained, cross-platform tool that mirrors the console workflow, LAMMPS-GUI removes this barrier, letting instructors teach one interface on all platforms and focus on LAMMPS and molecular dynamics itself.

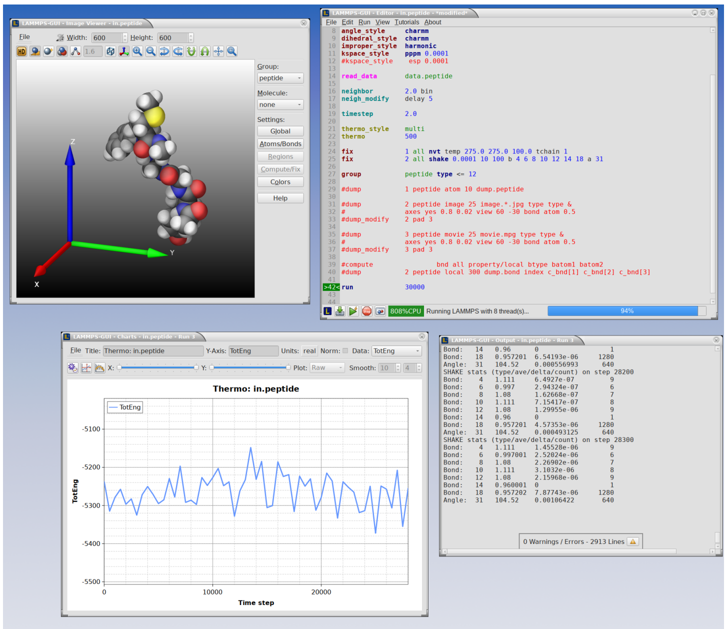

Figure 1: A LAMMPS-GUI session with a simulation in flight. The editor shows syntax highlighting, line numbers, and a marker on the current input line; the status bar shows CPU utilization and run progress; the Image Viewer shows the starting geometry; the Output window the screen output; and the Charts window a live plot of a thermodynamic column.

State of the field

Other approaches to making LAMMPS more accessible exist, but they target different needs. Commercial molecular-modeling packages provide structure editors that export LAMMPS inputs and re-import results; Atomify compiles a modified LAMMPS to WebAssembly and runs simulations and visualization inside a web browser (Hafreager, 2025); and dedicated visualization programs such as OVITO (Stukowski, 2010) and VMD (Humphrey et al., 1996) render trajectories with a breadth and quality that LAMMPS-GUI does not attempt to match. None of these, however, integrates input editing, live simulation, output monitoring, plotting, and visualization into one application that follows the same edit-run-observe loop as the standalone executable. LAMMPS-GUI fills that gap. It also imports tutorial materials, for example from the official LAMMPS tutorials (Gravelle et al., 2025), and has been used in that role at workshops. Although aimed primarily at beginners, features such as rapid prototyping of input decks, debugging failing inputs by visualizing selected components, and interactive construction of reproducible `dump image` commands also make it useful to experienced researchers.

Software design and functionality

LAMMPS-GUI follows an object-oriented design in which a central window coordinates largely self-contained components, all access to LAMMPS being funneled through a single adapter class. Key capabilities include:

- **Editing.** A LAMMPS-aware editor with syntax highlighting, per-command-category auto-completion, in-place documentation lookup, block comment/uncomment, and command reformatting.
- **Running.** Execution of the editor buffer through the library interface with no intermediate file I/O, a progress bar with estimated completion, support for the GPU, INTEL,

KOKKOS, and OPENMP accelerator packages, and recovery from simulation errors – reported as catchable exceptions – without crashing.

- **Output and charts.** An output window that highlights warnings and errors and makes documentation URLs clickable, and a charts window that plots thermodynamic data read directly from the running simulation (not scraped from text) or from external files, with Savitzky-Golay smoothing (Savitzky & Golay, 1964) and post-processing such as autocorrelation, polynomial, Birch-Murnaghan equation-of-state, and nonlinear fits.
- **Visualization.** A snapshot image viewer that builds high-quality renderings via the LAMMPS `dump image` command, with interactive controls and the ability to copy the generated command back into the script for reproducible figures, plus a slide-show viewer for image sequences exportable as movies.
- **Tutorials and convenience.** A guided wizard that downloads lesson inputs from several tutorial collections, a tabbed preferences dialog, hand-off to OVITO and VMD, and command-line flags exposing the charting and image viewers as standalone utilities.

A recurring theme is the co-evolution of LAMMPS-GUI and LAMMPS: front-end needs drove improvements to the engine and its library interface – catchable exceptions, a locked cache of live thermodynamic data, and greatly expanded snapshot rendering, collected into a dedicated GRAPHICS package (Kohlmeyer & Berger, 2025). This rendering capability deliberately draws on the kind of functionality long provided by dedicated molecular visualization tools, but makes it available directly on a running simulation rather than only as post-processing on saved trajectory files. In its default *plugin mode*, LAMMPS-GUI loads the shared library at run time with no link-time dependency, so one binary can pair with – or download – different LAMMPS builds. The project is documented online (Kohlmeyer, 2026) and maintained with an automated test suite and continuous-integration builds on all platforms.

AI usage disclosure

The design and architecture of LAMMPS-GUI were developed without the use of generative AI. AI-based coding assistants (GitHub Copilot and Claude Code) were used more recently to support the refactoring and modularization of the existing code base, and to assist with editing this manuscript. All AI-generated suggestions were reviewed, tested, and revised by the author, who takes full responsibility for the software and for the content of this paper.

Acknowledgements

Financial support by Sandia National Laboratories under PO 2149742 and PO 2407526, by the US National Science Foundation via Major Research Infrastructure grants number 1625061 and 2216289, and by CCDC-ARL under Cooperative Agreement Number W911NF-21-2-0007 is gratefully acknowledged. The author thanks the LAMMPS developers and the authors of the LAMMPS tutorials for their feedback and for motivating many of the features described here.

References

FrantzDale, B., Plimpton, S. J., & Shephard, M. S. (2010). Software components for parallel multiscale simulation: An example with LAMMPS. *Engineering with Computers*, *26*, 205–211. https://doi.org/10.1007/s00366-009-0156-z

Gravelle, S., Alvares, C. M. S., Gissinger, J. R., & Kohlmeyer, A. (2025). A set of tutorials for the LAMMPS simulation package. *Living Journal of Computational Molecular Science*, *6*(1), 3037. https://doi.org/10.33011/livecoms.6.1.3037

Hafreager, A. (2025). *Atomify*. https://andeplane.github.io/atomify

Humphrey, W., Dalke, A., & Schulten, K. (1996). VMD: Visual molecular dynamics. *Journal of Molecular Graphics*, *14*(1), 33–38. https://doi.org/10.1016/0263-7855(96)00018-5

Kohlmeyer, A. (2026). *LAMMPS-GUI documentation*. https://lammps-gui.lammps.org/

Kohlmeyer, A., & Berger, R. (2025). LAMMPS: A case study for applying modern software engineering to an established research software package. *US Research Software Engineering Conference (US-RSE'25), Philadelphia, PA, USA*. https://doi.org/10.5281/zenodo.17410115

Savitzky, A., & Golay, M. J. E. (1964). Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, *36*(8), 1627–1639. https://doi.org/10.1021/ac60214a047

Stukowski, A. (2010). Visualization and analysis of atomistic simulation data with OVITO – the open visualization tool. *Modelling and Simulation in Materials Science and Engineering*, *18*(1), 015012. https://doi.org/10.1088/0965-0393/18/1/015012

The Qt Company. (2026). *Qt cross-platform framework*. https://www.qt.io/

Thompson, A. P., Aktulga, H. M., Berger, R., Bolintineanu, D. S., Brown, W. M., Crozier, P. S., Veld, P. J. in't, Kohlmeyer, A., Moore, S. G., Nguyen, T. D., Shan, R., Stevens, M. J., Tranchida, J., Trott, C., & Plimpton, S. J. (2022). LAMMPS – a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Computer Physics Communications*, *271*, 108171. https://doi.org/10.1016/j.cpc.2021.108171